IOWA STATE UNIVERSITY
Department of Chemical and Biological Engineering

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Colocalization-Confirmed Transcriptome-Wide Association Study of Ankylosing Spondylitis to Prioritize Tissue-Specific Drug Targets

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Introduction

Ankylosing spondylitis (AS) is a chronic inflammatory arthritis of the spine and sacroiliac joints affecting approximately 0.1–0.5% of the population, with onset typically before age 30. While genome-wide association studies (GWAS) have identified over 100 risk loci, the majority fall in non-coding regions, making it difficult to pinpoint causal genes and actionable drug targets.

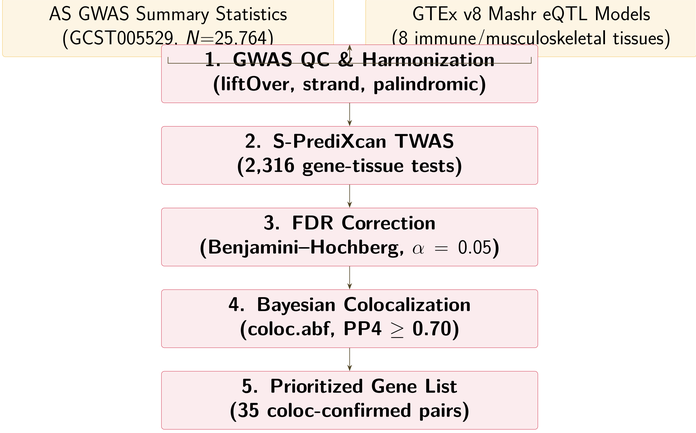
Transcriptome-wide association studies (TWAS) integrate GWAS with gene expression prediction models to identify genes whose genetically regulated expression is associated with disease risk. However, TWAS hits can arise from *linkage disequilibrium (LD) contamination* rather than true shared causal variants.

Bayesian colocalization (coloc) tests whether a GWAS signal and an eQTL signal share the same causal variant, distinguishing genuine tissue-specific regulatory effects from LD artifacts.

Key Question

Which genes have genetically regulated expression that **causally colocalizes** with AS risk variants, and in which tissues?

Methods



GWAS QC: 123,941 input SNPs → liftOver hg38 → 14,032 palindromic SNPs removed → 109,893 harmonized variants.

Tissues: Whole Blood, Spleen, Lung, Colon Sigmoid, Colon Transverse, Small Intestine, Muscle Skeletal, Adipose Subcutaneous.

TWAS Results

S-PrediXcan tested **2,316 gene-tissue pairs** across 8 GTEx v8 tissues. After Benjamini–Hochberg FDR correction ($\alpha = 0.05$), **248 significant associations** were identified, spanning genes on multiple chromosomes.

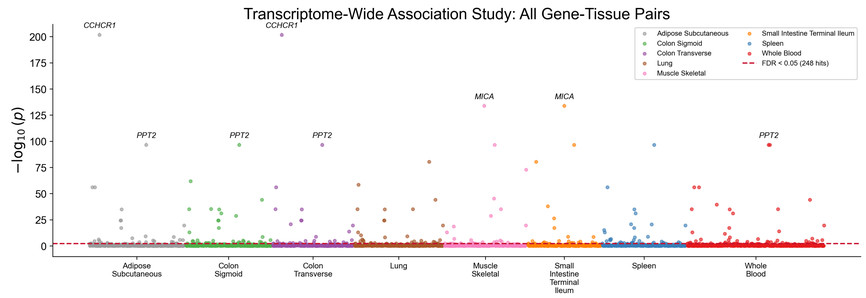


Figure. Manhattan-style plot of TWAS $-\log_{10}(p)$ across 8 tissues. Top gene symbols annotated. Dashed red line: FDR < 0.05.

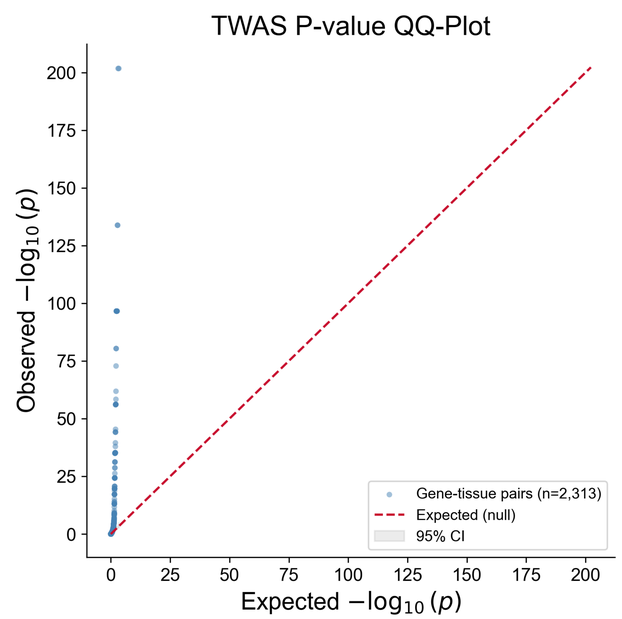
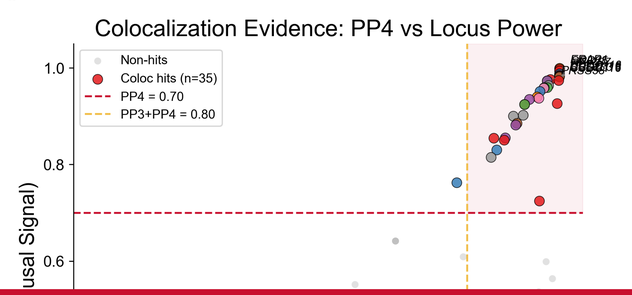


Figure. QQ-plot showing strong departure from the null, consistent with polygenic architecture. Grey band: 95% CI.

Colocalization Results

Of the 248 TWAS hits, Bayesian colocalization (coloc.abf) identified **35 gene-tissue pairs (18 unique genes)** with PP4 ≥ 0.70, confirming a shared causal variant between GWAS and eQTL signals.



Top Colocalized Genes

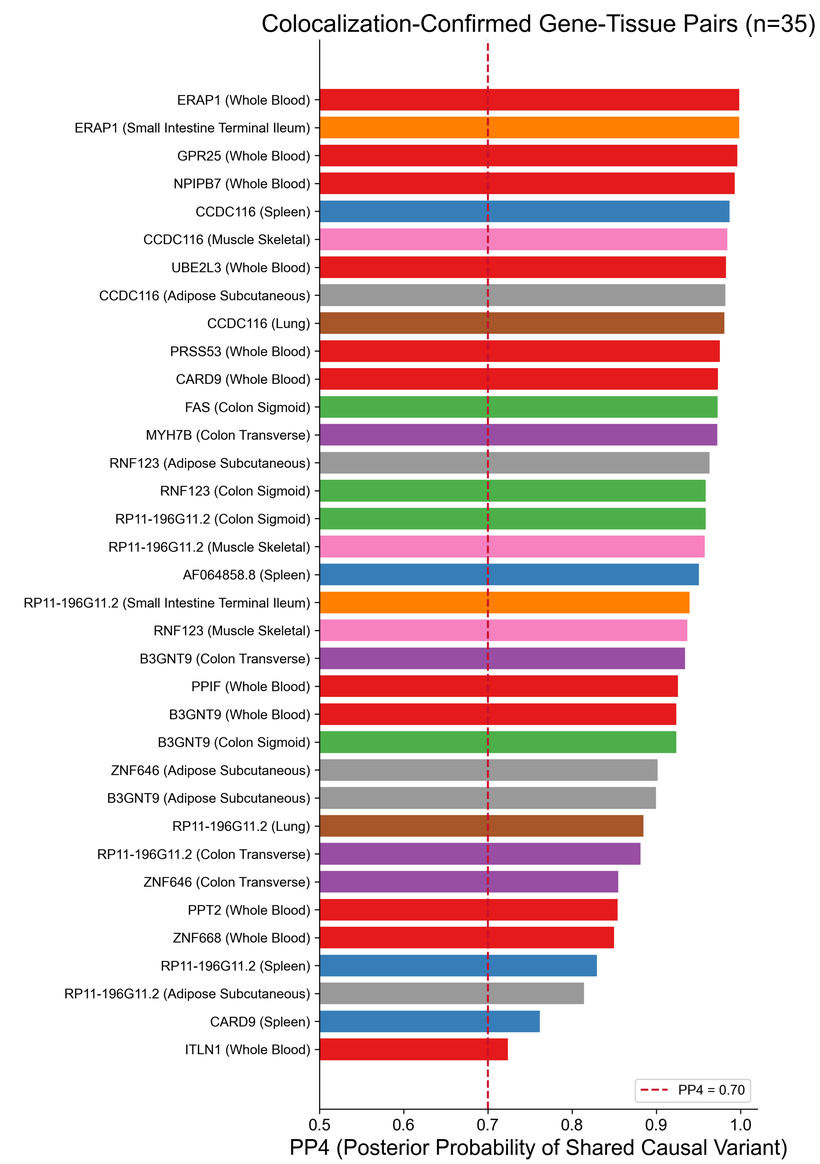


Figure. All 35 coloc-confirmed gene-tissue pairs ranked by PP4. Color indicates tissue. Dashed red line: PP4 = 0.70 threshold.

Gene	Top Tissue	PP4	z-score	FDR p
ERAP1	Whole Blood	0.999	13.28	2.5e-38
GPR25	Whole Blood	0.997	−7.03	6.0e-11
NPIPB7	Whole Blood	0.993	−5.16	5.9e-06
CCDC116	Spleen	0.988	−5.16	5.9e-06
UBE2L3	Whole Blood	0.983	5.32	2.5e-06
CARD9	Whole Blood	0.974	4.94	1.7e-05
FAS	Colon Sigmoid	0.973	4.42	1.9e-04
MYH7B	Colon Transverse	0.973	4.42	1.9e-04
RNF123	Adipose Subcut.	0.963	4.47	1.5e-04
PRSS53	Whole Blood	0.976	−3.68	3.3e-03

Discussion